FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Puerperium & Lactation

Lecture | Week 2 · Labour, delivery & the puerperium | Obstetrics | Dr Kaduyu

The puerperium is the 6 weeks (42 days) after delivery of the placenta, when the body returns to its non-pregnant state and lactation is established. Good postnatal care (at least 4 contacts, per WHO 2022) catches the killers of this period (PPH, sepsis, pre-eclampsia, VTE and suicide) and supports breastfeeding and postpartum family planning.

Must know before the ward round

1. Puerperium = from delivery of the placenta to 6 weeks (42 days). The uterus falls from about 1 kg to about 60 g and is no longer palpable abdominally by about day 10–14.
2. Lochia: rubra (days 1–4), serosa (about days 4–10), alba (about day 10 to 4–6 weeks). Offensive or persistently red lochia suggests infection or retained products.
3. WHO 2022: stay at least 24 h after vaginal birth and have at least 4 postnatal contacts: within 24 h, day 3 (48–72 h), days 7–14, and 6 weeks.
4. Postnatal check head to toe (BUBBLE-HE): Breasts, Uterus, Bladder, Bowels, Lochia, Episiotomy/wound, legs (calves), Emotions, plus temperature, pulse and BP every time.
5. She must pass urine within 6 h of birth. A high, deviated, 'boggy' fundus is often a full bladder; residual over about 500 mL → indwelling catheter for 24–48 h.
6. Puerperal fever: 7 Ws + malaria. Always do a malaria RDT and examine uterus, lochia, wound, breasts, chest, urine and calves before choosing antibiotics. WBC up to 20–25 × 10^9/L can be normal in the first days.
7. Puerperal sepsis: ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly until afebrile for 48 h. Always look for retained products, and test for malaria.
8. LAM gives over 98% protection only if all three hold: amenorrhoea, fully or nearly fully breastfeeding day and night, and baby under 6 months.
9. Combined oral contraceptives are category 4 if breastfeeding and under 6 weeks postpartum. Implants and POPs can start immediately; a postpartum IUD goes in within 48 h or after 4 weeks.
10. Postpartum blues (days 3–10, self-limiting) vs depression (≥2 weeks, screen with EPDS) vs psychosis (first 2 weeks, an emergency with risk of suicide and infanticide).
11. EPDS: ≥10 possible and ≥13 probable depression; any positive answer on item 10 (self-harm) needs same-day assessment. Sertraline is the preferred antidepressant in breastfeeding.
12. Mastitis: keep breastfeeding and emptying the breast; cloxacillin/flucloxacillin 500 mg orally 6-hourly for 10–14 days; an abscess needs drainage.

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1. Definitions and Key Concepts

The **puerperium** is the period from the **delivery of the placenta until 6 weeks (42 days)** after birth. During it the genital tract **involutes**, pregnancy's physiological changes reverse, and **lactation** is established.

It matters because **most maternal deaths occur after birth**, and most of those in the **first 24–48 hours**: PPH, then sepsis, hypertensive disorders, VTE and, later, **suicide**. It is also the time when **newborn deaths** cluster, when breastfeeding succeeds or fails, and when a woman who doesn't want another pregnancy soon needs **contraception** (ovulation can return from about **3–4 weeks** in a woman who isn't breastfeeding).

> **CLINICAL PEARL**
> The one thing never to forget: **"the woman is still a patient after the baby is born."** Check her vital signs, uterus, lochia, perineum, bladder, breasts, legs and mood, not just the baby.

Term	Definition
Puerperium	From delivery of the placenta to **6 weeks (42 days)** postpartum. **Immediate**: first 24 h; **early**: up to day 7; **remote/late**: up to 6 weeks
Involution	Return of the uterus (and genital tract) to its non-pregnant size and state
Lochia	Vaginal discharge after birth: blood, decidua, mucus, bacteria
Puerperal pyrexia (classic UK definition)	Temperature **≥38 °C** during the first **14 days** after birth (older definition: on any 2 of the first 10 days, excluding the first 24 h)
Puerperal sepsis (WHO)	Infection of the genital tract occurring between rupture of membranes or labour and the **42nd day postpartum**, with fever plus ≥1 of: **pelvic pain, abnormal vaginal discharge, abnormal smell of discharge, delayed uterine involution**
Maternal sepsis (WHO 2017)	**Life-threatening organ dysfunction** from infection during pregnancy, childbirth, post-abortion or the postpartum period
Lactational amenorrhoea method (LAM)	Temporary contraception from breastfeeding (see section 6E)
Exclusive breastfeeding	Only breast milk (no water, other liquids or foods) except medicines, vitamins, ORS, for **6 months**

2. Burden and Uganda Context

- Globally, **most maternal deaths and a large share of newborn deaths happen after birth**, clustered in the **first day and the first week**.
- **Postnatal care (PNC) coverage in Uganda lags behind ANC and facility delivery**: many women leave within hours of a normal birth and never return until the baby's first immunisation.
- **Puerperal sepsis** is a leading cause of maternal death in some Ugandan tertiary hospitals (**about 31% of deaths** in one tertiary-hospital series **Verify** *the 2016 BMC Pregnancy and Childbirth series appears to come from Mbarara Regional Referral Hospital, not Mulago as previously stated*). Risk factors are common locally: prolonged labour, multiple VEs, PROM, home births, CS, anaemia, HIV and poor hygiene.
- **Postpartum depression** is common in Uganda (studies report about 10–40% depending on the setting and tool), and usually undiagnosed.
- **Breastfeeding** is almost universal in Uganda, but **exclusive breastfeeding to 6 months** is achieved by only about two-thirds of infants, and **early initiation (within 1 hour)** is often missed after CS.

- **Unmet need for family planning** is highest in the postpartum year, and short birth intervals raise maternal and child mortality.

> **UGANDA CONTEXT**
> Ugandan health workers use the **"6 hours, 6 days, 6 weeks"** postnatal schedule, aligned with WHO's four contacts (within 24 h, day 3, days 7–14, 6 weeks). The **Essential Maternal and Newborn Clinical Care Guidelines (MoH 2022)** and **UCG 2023** recommend that women stay in the facility for **at least 24 hours** after a normal delivery. In practice, beds are scarce and many are discharged at 6–12 hours. That is why **danger-sign counselling at discharge** is critical.

3. Principles and Framework

Three processes run side by side after the placenta is delivered: the **genital tract involutes**, the **other systems reverse the changes of pregnancy**, and **lactation** is established. Each normal change has a matching way of going wrong, which is what you look for at every postnatal contact.

The genital tract

Structure	Change
Uterus	Weighs ~1000 g at delivery → ~500 g at 1 week → ~300 g at 2 weeks → **~50–100 g by 6 weeks**. The fundus is at the **umbilicus** just after delivery (firm, "cricket ball"), descends about **1 cm (one finger-breadth) per day**, and is **no longer palpable abdominally by day 10–14**. Mechanism: **autolysis** of myometrial cells (proteolytic enzymes reduce cell size, not number) and **ischaemia** from contraction
Placental site	Thrombosed vessels are gradually excluded by endometrial regrowth from the margins, without scarring. It is the last area to heal (about 6 weeks). Failure = **subinvolution** of the placental site (secondary PPH)
Endometrium	Regenerates from the basal layer by about day 10 (except the placental site, about 6 weeks)
Cervix	Soft and patulous after delivery; closes to admit one finger by about **the end of week 1**; the external os becomes a transverse slit (parous os)
Vagina / perineum	Oedema and bruising settle; rugae return from week 3; tears heal in about 1–2 weeks. Hypo-oestrogenic dryness while breastfeeding
Afterpains	Painful contractions during breastfeeding (oxytocin), worse in multiparas; they help involution

Lochia:

Type	When	Appearance
Rubra	Days 1–4	Red (blood, decidua)
Serosa	About days 4–10	Pink-brown (less blood, more serum and leucocytes)
Alba	About day 10 to 4–6 weeks	Yellow-white (leucocytes, mucus)

RED FLAG

Offensive lochia, **persistent red lochia beyond about 2 weeks**, a return of heavy bleeding, or a **uterus that is bigger, softer or more tender than expected** (subinvolution) point to **infection or retained products**.

Other systems

System	Change and clinical relevance
Cardiovascular	Cardiac output **rises immediately after delivery** (autotransfusion of about 500 mL from the contracted uterus and relief of caval compression), then falls towards normal over **2–6 weeks** (up to 12 weeks for full remodelling). **This is when women with heart disease (e.g. mitral stenosis) decompensate into pulmonary oedema.**
Blood	Physiological **diuresis on days 2–5** removes the extra plasma volume. Clotting factors and fibrinogen stay high, so the woman is **hypercoagulable for about 6 weeks**, and **VTE risk is highest in the first weeks** (especially after CS). Leucocytosis up to 20–25 × 10^9/L is normal in the first days
Urinary	Bladder oedema and reduced sensation after labour/epidural lead to **urinary retention** (overflow). Dilated ureters return to normal by 6–8 weeks. **Always check that she has passed urine within 6 h of birth**
Gastrointestinal	Constipation (progesterone effect, fear of pain from the perineum, opioids, dehydration)
Endocrine	Placental hormones (hPL, oestrogen, progesterone, hCG) fall rapidly (hCG is undetectable in about 2–4 weeks). **Prolactin stays high while breastfeeding**. **Ovulation** returns in non-breastfeeding women as early as **day 25** (mean about 45 days), and is suppressed by frequent breastfeeding
BP	Often rises on **days 3–6** postpartum; pre-eclampsia (and eclampsia) can **first appear postpartum** (up to 6 weeks)
Abdominal wall / skin	Rectus diastasis, striae fade, linea nigra fades
Weight	About 5–6 kg is lost at delivery and a further 2–4 kg with diuresis
Psychological	Rapid hormonal shifts, sleep deprivation and the new role make this a vulnerable period for mood disorders

Lactation physiology

1. **Mammogenesis** (pregnancy): **oestrogen** → duct growth; **progesterone** → lobulo-alveolar growth; plus **prolactin, hPL**, cortisol and insulin.
2. **Lactogenesis I** (from mid-pregnancy): the alveoli can make **colostrum**, but high **progesterone blocks full milk secretion**.
3. **Lactogenesis II** ("milk coming in", about **day 2–4**): triggered by the **fall in progesterone after delivery of the placenta**, with prolactin already high. **Retained placental fragments** (progesterone) or **severe PPH (Sheehan's)** delay or prevent it; so can diabetes, obesity, stress and CS.
4. **Galactopoiesis** (maintenance): requires **prolactin** (from the anterior pituitary, released in response to **suckling** and higher at night) and **regular removal of milk**. A local protein, **feedback inhibitor of lactation (FIL)**, accumulates in a full breast and **reduces production**. So **"supply follows demand"**: the more the baby removes, the more is made.
5. **Milk ejection (let-down) reflex**: suckling → **oxytocin** (posterior pituitary) → contraction of **myoepithelial cells** → milk flows. It can be **conditioned** (the baby's cry) and is **inhibited by pain, anxiety, embarrassment**. Oxytocin also causes **afterpains** and helps involution.

Milk	Features
Colostrum (days 1–3)	Small volume (a few mL per feed, matching the newborn's small stomach), yellow, **rich in IgA, lactoferrin, leucocytes, protein, vitamin A**; laxative (clears meconium); "the first immunisation"
Transitional (days 4–14)	Rising volume
Mature milk	**Foremilk** (watery, lactose-rich, quenches thirst) then **hindmilk** (fat-rich, calorie-dense). Let the baby finish one breast before switching

Why breastfeeding contraception (LAM) works: frequent suckling → high prolactin and neural signals → **suppressed GnRH pulsatility** → low LH/FSH → no follicular development → **anovulation and amenorrhoea**. Long gaps between feeds (especially at night) or supplementary feeds weaken this.



4. Application in Practice

On the postnatal ward you will do two kinds of workup: the **routine postnatal check** (every woman, every contact) and the **assessment of an unwell postnatal woman** (fever, bleeding, headache, breathlessness, low mood). They use the same steps; the unwell woman simply gets Step 0 first and more of Steps 3 and 4.

Step 0: First 5 minutes

For **any postnatal woman who looks unwell or has a danger sign**:

1. **ABC and full vital signs**: temperature, pulse, BP, RR, SpO_2, conscious level. Plot them on the **MEOWS/early warning chart**.
2. **Sepsis screen**: fever ≥38 °C or temperature <36 °C, pulse >100, **RR ≥22, SBP ≤100, altered mentation**. Any two, with a likely source → treat as sepsis now: cultures, malaria test, **IV antibiotics within 1 hour**, fluids.
3. **Bleeding check**: pad and lochia (look and smell), **shock index** (≥0.9 abnormal), feel the fundus. Heavy bleeding → secondary PPH pathway.
4. **BP ≥160/110, severe headache, visual disturbance or a fit** → postpartum (pre-)eclampsia: $MgSO_4$, antihypertensive (see the pre-eclampsia notes).
5. **Sudden breathlessness, chest pain, hypoxia** → PE or peripartum cardiomyopathy/pulmonary oedema: oxygen, sit up, senior review.
6. **Mental state**: confusion, hallucinations, not sleeping, talk of harming herself or the baby → **do not leave her alone with the baby**; exclude organic causes (glucose, malaria, sepsis, BP).
7. **Call for help** early; IV access and bloods (FBC, cultures, malaria test, group and cross-match if bleeding).

RED FLAG

A postnatal woman who is **tachycardic, breathless and "just not right" on day 1–2, with little fever and no smelly lochia**, may have **group A streptococcal sepsis**. It can kill within hours: IV antibiotics, fluids and senior review immediately.

Step 1: Focused history

Ask	Why: the answer that changes the plan
Day postpartum; date, mode and place of birth; length of labour; **hours of ruptured membranes**; number of VEs; manual removal, tear or episiotomy; CS indication; **placenta complete?**	Sepsis risk (prolonged labour, PROM, multiple VEs, CS), retained products, wound problems; CS → wound, VTE
Blood loss at delivery and Hb	Anaemia; Sheehan's syndrome (failed lactation after major PPH)
Bleeding: amount, colour and **smell** of lochia; clots; return to red	Secondary PPH, infection
Fever, chills, rigors, pain (abdomen, perineum, wound, breast, loin, calf, chest)	Finds the "W": endometritis, malaria, UTI, mastitis, wound infection
Headache, visual disturbance, epigastric pain, swelling	**Postpartum pre-eclampsia**
Breathlessness, chest pain, calf pain or swelling	**PE / DVT**, peripartum cardiomyopathy
Bladder: passed urine since delivery (**within 6 h**)? Pain, frequency, **continuous leakage**	Retention, UTI, **obstetric fistula** (VVF after obstructed labour)
Bowels: opened? Constipation, faecal or flatus incontinence	Missed 3rd/4th-degree tear, RVF
Perineal pain	Haematoma, infection, wound breakdown
Breastfeeding: how often (**8–12 feeds/24 h**), attachment, nipple pain, breast lumps or redness, baby's wet nappies	Attachment problems, mastitis, low intake
Mood: sleep, anxiety, tearfulness, loss of interest (**Whooley questions**), **thoughts of harming herself or the baby**, bonding	Blues, depression, psychosis
Support at home, domestic violence	Risk for depression and neglect
HIV status, ART adherence, baby's prophylaxis; **Rh status**	eMTCT; anti-D within 72 h
Medical conditions: hypertension, diabetes, heart disease, anaemia, TB, sickle cell, epilepsy	Follow-up and drug changes after delivery
Contraception plans, return of sexual activity, **fully breastfeeding?**	PPFP: ovulation comes before the first period; LAM or another method today

Step 2: Focused examination

Follow the order of Hutchison's Clinical Methods (24th edition); the system examination is the **BUBBLE-HE** sequence.

Introduction, consent and preparation

- **Introduce yourself**, explain the check and obtain **consent**; offer a **chaperone** for the breast and perineal examination.
- **Hand hygiene** (and gloves for the perineum and lochia).
- **Position and exposure**: lying comfortably, semi-recumbent; expose the breasts, then the abdomen, then the perineum in turn, keeping the rest covered. Ask her to **empty her bladder** first unless you are assessing retention.

General inspection and vital signs

- Well or unwell, pallor (anaemia), distress, **mood and affect**, how she handles the baby.
- **Temperature, pulse, BP, RR, SpO_2**, plotted on the early warning chart (MEOWS).

Hands, face and neck

- Hands: pallor, capillary refill, warm or cold peripheries (sepsis, bleeding).
- Face: conjunctival pallor, jaundice; facial oedema (pre-eclampsia).
- Neck: thyroid (postpartum thyroiditis can present at 2–6 months).

The postnatal system examination (BUBBLE-HE)

Letter	Examine	Normal	Abnormal (think of)
Vitals	T, P, BP, RR, SpO_2	T <38 °C, P <100, BP <140/90	Sepsis, bleeding, pre-eclampsia, PE
Breasts	Inspect and palpate; watch a feed	Soft to full, nipples intact	Engorgement (bilateral), mastitis (unilateral red wedge), abscess (fluctuant), cracked nipples
Uterus	Fundal height, consistency, tenderness	Firm, central, non-tender; about 1 cm (a finger-breadth) lower each day; impalpable by day 10–14	Soft, bulky, tender, higher than expected = subinvolution, endometritis, retained products
Bladder	Suprapubic dullness, fundus position	Voided within 6 h; not palpable	Palpable bladder, fundus pushed up and to one side = retention
Bowels	Ask; abdominal distension, bowel sounds (after CS)	Opened by day 2–3	Ileus, constipation, incontinence
Lochia	Look at the pad and **smell** it	Rubra → serosa → alba; not offensive	Offensive, heavy, persistently red
Episiotomy / wound	Perineum and CS wound	Clean, edges together, mild swelling	Redness, pus, gaping, haematoma, dehiscence
H (legs)	Calves, oedema	No swelling or tenderness	Unilateral swelling, tenderness (DVT). Homans' sign is unreliable: do not rely on it
Emotions	Mood, affect, bonding, sleep	Blues: tearful but coping	Low mood ≥2 weeks, suicidal thoughts, psychotic features

Add a **chest** exam (crackles: pneumonia, aspiration, pulmonary oedema) and **heart** exam (murmur, gallop: cardiac disease, peripartum cardiomyopathy) in any unwell woman. **Speculum/VE** only if indicated (retained products, fistula, wound breakdown).

Completing the examination

- Cover her, **thank her**, **hand hygiene**.
- **Summarise**: day postpartum, vitals, involution, lochia, wound, breasts and feeding, bladder, mood, and an impression (normal puerperium, or the problem and plan).

Document: day postpartum; vitals; fundal height in finger-breadths or cm below the umbilicus; lochia type and smell; wound; breasts and feed observed; voided (yes/no, time); mood; FP method chosen; and negatives such as "calves soft, non-tender", "lochia not offensive", "no suicidal ideation".

Step 3: Bedside tests

Test	When	What it tells you
Temperature, pulse, BP, RR (early warning chart)	Every contact	Sepsis, bleeding, postpartum pre-eclampsia
Malaria RDT	Any fever	Malaria, the commonest non-genital cause of puerperal fever in Uganda
Urine dipstick	Fever, dysuria, high BP	Nitrites/leucocytes (UTI); protein (pre-eclampsia)
HemoCue / point-of-care Hb	After PPH, pallor, before discharge if anaemic	Anaemia (treat, or transfuse if Hb <7 g/dL or symptomatic)
Random blood glucose	Confused, drowsy, diabetic, fitting	Hypoglycaemia, hyperglycaemia
Bladder volume (in-out catheter, or bladder scan where available)	Not voided within 6 h, palpable bladder	Residual >500 mL → indwelling catheter 24–48 h
Observe a breastfeed	Every contact in the first week; any breast problem	Attachment (4 signs), effective suckling
Baby's weight and wet nappies	Feeding concerns	≥6 wet nappies/day after day 5 = adequate intake
Whooley questions / EPDS	At contacts; 6 weeks; any mood concern	EPDS ≥10 possible, ≥13 probable depression; item 10 = safety
Point-of-care ultrasound (skilled)	Bleeding, fever with a bulky uterus	Retained products, collections
HIV rapid test (if negative or unknown)	Per national retesting schedule	Seroconversion during breastfeeding

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Full blood count	Fever, bleeding, pallor	Hb <10–11 g/dL anaemia; WBC **up to 20–25 × 10^9/L can be normal** in the first days, so trend it and look for left shift	Iron vs transfusion; supports sepsis only with the clinical picture. HC IV and above
Blood smear for malaria	Fever, RDT unavailable or to quantify	Parasitaemia	Oral ACT (uncomplicated) or IV artesunate (severe)
Blood cultures (before antibiotics)	Sepsis	GAS, *E. coli*, anaerobes, *Staph. aureus*	Narrow or change antibiotics. RRH/NRH/private labs
HVS / endocervical swab	Endometritis, offensive lochia	Organisms	Guides antibiotics; gonorrhoea/chlamydia → treat partner
Urine microscopy and culture (MSU)	Dysuria, loin pain, catheter	Pyuria, organism	Pyelonephritis regimen
Wound swab / pus	Wound infection, abscess	Organism (often *Staph.*)	Antibiotic choice; drainage
Pelvic ultrasound	Bulky tender uterus, secondary PPH, persistent fever	Retained products, pelvic collection, haematoma	**MVA** (after antibiotics started), drainage or laparotomy. HC IV (if sonographer) and above

Test	When	Expected / abnormal finding	How it changes management
Breast ultrasound	Lump not settling in 48 h on antibiotics	Abscess cavity	Needle aspiration vs incision and drainage
Leg Doppler ultrasound	Unilateral leg swelling	Non-compressible vein (DVT)	Therapeutic LMWH/heparin, then warfarin (compatible with breastfeeding). Hospitals
Creatinine, LFTs, platelets	Postpartum pre-eclampsia, sepsis	Raised creatinine/transaminases, low platelets	HELLP, AKI; $MgSO_4$, adjust gentamicin
Lactate	Suspected septic shock	>2 mmol/L (>4 severe)	Fluids, vasopressors, HDU/ICU. NRH/ICU mainly
Chest X-ray, ECG, echo	Breathlessness, suspected PPCM, pneumonia	Cardiomegaly, pulmonary oedema, consolidation	Heart-failure treatment, cardiology; antibiotics. Hospitals (echo at RRH/NRH)
β-hCG	Bleeding persisting beyond 6 weeks or after evacuation	Raised or plateauing	Suspect gestational trophoblastic neoplasia
TSH / thyroid function	Fatigue, palpitations, low mood at 2–6 months	Postpartum thyroiditis	Symptomatic treatment; check before labelling as depression. RRH/NRH/private
Viral load (HIV)	Per national schedule during breastfeeding	Not suppressed	Adherence support; enhanced infant prophylaxis

Principles for interpreting tests in the puerperium

1. **Most postnatal diagnoses are clinical.** A good BUBBLE-HE examination, a temperature chart and a malaria RDT answer most questions; tests confirm and guide treatment.
2. **Interpret results against postnatal physiology**: the WBC is normally raised in the first days, fibrinogen and D-dimer stay high (so D-dimer cannot exclude VTE), and BP often rises on days 3–6.
3. **Take cultures before antibiotics but never delay antibiotics for them**: in suspected sepsis, antibiotics go in within 1 hour.

Step 5: Putting it together

Model one-line summary

> "A 24-year-old para 2, day 4 after an emergency CS for obstructed labour with 20 h of ruptured membranes, with fever 38.9 °C, pulse 112, RR 22, a tender uterus 2 cm above the umbilicus and offensive lochia; malaria RDT negative: puerperal sepsis from post-CS endometritis, retained products to be excluded."

Problem list

1. Puerperal sepsis (endometritis), on IV ampicillin, gentamicin and metronidazole.
2. Possible retained products (subinvolution).
3. Post-CS: wound, VTE risk (thromboprophylaxis).
4. Breastfeeding while unwell: support, express if needed.
5. PPFP not yet chosen.

Working diagnosis: *post-caesarean endometritis with sepsis; exclude retained products.*

EXAM FAVOURITE

Worked example 1: two postnatal checks.

- **Mother A, day 3 after SVD**: T 36.8 °C, P 84, BP 122/78. Fundus **three finger-breadths below the umbilicus**, firm, central, non-tender. Lochia rubra, moderate, not offensive. Perineum: second-degree repair intact. Breasts full, good attachment observed. Tearful at times but sleeping when the baby sleeps. **Interpretation**: normal involution, normal lochia, **postpartum blues**; reassure, danger signs, iron and folic acid, PPFP, next contact at days 7–14.
- **Mother B, day 7 after a home birth**: T 38.4 °C, P 108. Fundus **at the umbilicus**, soft and tender (it should be well below it by now). Lochia still red and **offensive**. **Interpretation**: **subinvolution with endometritis**, likely retained products. Plan: malaria test, cultures, IV antibiotics, pelvic ultrasound, MVA after antibiotics have started.

EXAM FAVOURITE

Worked example 2: a "high uterus" on day 1. Six hours after SVD, the fundus is **two finger-breadths above the umbilicus and deviated to the right**, and she has not passed urine. This is a **full bladder**, not atony. In-out catheter drains **700 mL** (>500 mL) → leave an **indwelling catheter for 24–48 h**, then a trial without catheter; check that she voids and measure the residual again. The fundus returns to the midline, below the umbilicus.

EXAM FAVOURITE

Worked example 3: interpreting the EPDS at 6 weeks. Score **14** (≥13 = probable depression), and she scores **1 on item 10** ("the thought of harming myself has occurred to me: hardly ever"). Action: **same-day risk assessment** by a clinician (plans, means, protective factors, risk to the baby), then psychological therapy ± **sertraline 50 mg daily**, review within 2 weeks, and refer to mental health services if the risk is significant. Check Hb and thyroid function if fatigue is prominent.

Model ward-round presentation

> "Mrs P, a 24-year-old para 2, is day 4 after an emergency CS at Mengo for obstructed labour after 18 hours in labour with membranes ruptured for 20 hours. She has had fever up to 38.9 °C since yesterday, lower abdominal pain and foul-smelling lochia. Pulse 112, BP 104/66, RR 22. The uterus is tender and 2 cm above the umbilicus (subinvolution), and the wound is clean. Malaria RDT negative; WBC 19 × 10^9/L. Impression: **puerperal sepsis from post-CS endometritis**, with retained products to be excluded. Plan: IV ampicillin, gentamicin and metronidazole (started 10:00), blood culture and HVS taken, pelvic ultrasound, fluids, catheter for urine output, continue breastfeeding with support, thromboprophylaxis, and review in 24 h. Offered an implant; she will decide after discussing with her husband."

5. Clinical Acumen

Thinking like a clinician

- **Fever in the puerperium: think by system.** Use the **"7 Ws"**: **W**omb (endometritis), **W**ind (chest infection/aspiration after GA), **W**ater (UTI/pyelonephritis), **W**ound (CS or perineum), **W**alking (DVT/PE), **W**eaning (breast engorgement/mastitis), **W**onder drugs (drug fever, transfusion reaction). **In Uganda add an eighth: "Where's the mosquito?" (malaria).** HIV-associated infections (TB, PCP) also belong here.
- **Timing clues**: day 1–2 fever with an ill mother and **no foul smell** → think **group A streptococcus** (can be fulminant) or aspiration; days 2–5 → endometritis, UTI; day 3–5 → engorgement (low-grade fever); days 5–10

→ wound infection, mastitis; any time → malaria, DVT.

- **The uterus is the clue**: tender, bulky and soft with offensive lochia means endometritis ± retained products. A **normal uterus with sepsis** makes you look elsewhere.
- **Fever not settling after 48–72 h of antibiotics** → pelvic abscess, infected haematoma, **septic pelvic thrombophlebitis**, resistant organisms, retained products, necrotising fasciitis, malaria, or a wrong diagnosis.
- **Postpartum headache**: not always "just tiredness". Check BP and urine (pre-eclampsia), think of post-dural-puncture headache (after spinal), cerebral venous sinus thrombosis (severe, seizures, focal signs) and meningitis.
- **Mood**: blues is **normal and transient**; symptoms lasting **more than 2 weeks** or severe symptoms are **not** blues. **Psychosis is an emergency.**
- **Common student mistakes**: not checking the uterus and lochia in a febrile woman; forgetting malaria; assuming breast pain means stopping breastfeeding; missing urinary retention (a "palpable uterus" that is actually a full bladder); forgetting contraception; not asking about mood.

EXAM FAVOURITE

"A woman on day 4 has a temperature of 38.5 °C. How will you approach her?" Answer with a **systematic search for the source** (7 Ws + malaria), the **examination** of breasts, chest, abdomen/uterus, wound, perineum, lochia and calves, the **investigations** (FBC, blood culture, malaria test, urinalysis/MSU, HVS, wound swab, ultrasound), and **empirical treatment** of the most likely source.

Red flags (maternal danger signs after birth)

RED FLAG

- **Heavy bleeding** (soaking more than 1 pad in an hour), passing large clots
- **Fever ≥38 °C**, rigors, foul-smelling lochia, severe abdominal pain
- **Severe headache, blurred vision, epigastric pain, convulsions**
- **Difficulty breathing, chest pain**; **painful swollen calf**
- **Tachycardia >100 with hypotension**, confusion (sepsis or bleeding)
- Perineal or wound pain getting worse, discharge from the wound
- **Thoughts of self-harm or harming the baby**, hallucinations, confusion, not sleeping for days
- Continuous leakage of urine or faeces

Scenarios: differential diagnosis

Puerperal fever:

Condition	Distinguishing features	Key investigation
Endometritis	Days 2–10; fever, lower abdominal pain, **tender bulky uterus, offensive lochia**; after CS, prolonged labour, PROM	FBC, blood culture, HVS; ultrasound for retained products
Malaria	Fever, chills, headache, joint pains; uterus and lochia normal	**RDT or blood smear**
UTI / pyelonephritis	Dysuria, frequency, loin pain, catheterisation	Urinalysis, MSU culture
Mastitis / breast abscess	Days 5–30; unilateral red, hot, tender wedge-shaped area; fluctuant lump = abscess	Clinical; ultrasound for abscess; milk culture if recurrent
Breast engorgement	Days 3–5; bilateral, hard, painful breasts; low-grade fever	Clinical
Wound infection (CS/episiotomy)	Days 4–10; redness, discharge, dehiscence	Wound swab

Condition	Distinguishing features	Key investigation
DVT / PE	Unilateral calf swelling or pain; tachycardia, dyspnoea, pleuritic chest pain	Doppler ultrasound; CTPA/V-Q (where available)
Chest infection	Cough, crackles, after GA (aspiration)	CXR
Septic pelvic thrombophlebitis	Spiking fever despite antibiotics, no clear source	Diagnosis of exclusion; CT
Pelvic abscess / infected haematoma	Persistent fever, pelvic mass, rectal fullness	Ultrasound

Postpartum mood disorders:

Condition	Onset / duration	Features	Action
Postpartum blues	Days 3–5 (peak), resolves by about day 10–14; very common (up to 50–80%)	Tearfulness, irritability, mood swings, anxiety; functioning preserved	Reassure, support, rest; review if lasting more than 2 weeks
Postpartum depression	Usually within the first weeks to months (up to 1 year); about 10–15% (often more in Uganda)	Low mood, anhedonia, guilt, poor sleep even when the baby sleeps, poor bonding, **suicidal thoughts**	**EPDS screen**, psychological therapy, antidepressant (sertraline), follow-up
Puerperal psychosis	Usually within the **first 2 weeks** (often days 3–10); about 1–2 per 1000	Confusion, agitation, **delusions, hallucinations**, mania, severe insomnia; **risk of suicide and infanticide**	**Emergency psychiatric admission**; do not leave her alone with the baby
Organic causes	Any	Delirium from sepsis, eclampsia, hypoglycaemia, thyroiditis, cerebral malaria, HIV/CNS infection	Exclude before labelling as psychiatric

6. Common Problems and Their Management

A. Routine postnatal care

The WHO 2022 schedule (Uganda: "6 hours, 6 days, 6 weeks"):

Contact	Timing	Key tasks
1	**Within 24 h** (stay in the facility **at least 24 h** after vaginal birth)	Vital signs and bleeding **every 15 min for the first 2 hours**, then at intervals; uterus, lochia, bladder (**passed urine within 6 h?**), perineum; initiate breastfeeding **within 1 hour**, skin-to-skin; newborn exam, vitamin K, eye care, cord care (chlorhexidine where indicated), BCG/OPV0; danger signs
2	**Day 3 (48–72 h)**	Check BP (postpartum pre-eclampsia peaks days 3–6), temperature, breasts and feeding, jaundice in the baby, perineum, mood
3	**Days 7–14**	Involution, wound healing, breastfeeding, mood (blues should be resolving), weight of baby
4	**6 weeks**	Full check: BP, Hb if anaemic, pelvic floor/continence, **family planning**, cervical cancer screening (VIA) if due, HIV retesting, **mood (EPDS)**, baby's immunisation (DPT-HepB-Hib 1, PCV1, OPV1, rotavirus) and growth

Every contact: counsel on **danger signs**, nutrition (extra meal a day while breastfeeding), rest, hygiene, **exclusive breastfeeding**, sexual health and **FP**, and **iron + folic acid** (continue for about 3 months after birth; ferrous sulphate 200 mg daily + folic acid).

After CS: early mobilisation, analgesia, wound care (remove dressing at 24–48 h, keep clean and dry), **thromboprophylaxis** if risk factors, breastfeeding support (positions that avoid the wound), avoid heavy lifting for about 6 weeks, and discuss interpregnancy interval (at least 18–24 months) and mode of next delivery.

Rh-negative mother with Rh-positive baby: **anti-D within 72 hours** of birth.

HIV-positive mother: continue **ART (TLD)** without interruption, check viral load, **infant prophylaxis** (per the Uganda consolidated HIV guideline), and **exclusive breastfeeding** for 6 months, continuing breastfeeding with complementary feeds while the mother is virally suppressed. Retest HIV-negative mothers during breastfeeding per the national schedule.

B. Perineal care

1. **Hygiene**: wash with clean water 2–3 times a day and after toileting; change pads every 4–6 h; front-to-back wiping.
2. **Analgesia**: **paracetamol 1 g orally every 6–8 h** (max 4 g/day) ± **ibuprofen 400 mg every 8 h** with food (safe in breastfeeding; avoid in severe pre-eclampsia with renal impairment or PPH with AKI); cold packs in the first 24–48 h; warm sitz baths afterwards.
3. **Prevent constipation**: fluids, fibre; lactulose 15 mL twice daily (routine after 3rd/4th-degree tears).
4. **Review** for infection or breakdown: redness, pus, gaping. Treat infection with antibiotics (e.g. amoxicillin-clavulanate or amoxicillin + metronidazole). Open, clean and allow **healing by secondary intention**, with resuturing later once clean.
5. **3rd/4th-degree tears (OASIS)**: broad-spectrum antibiotics, laxatives, physiotherapy/pelvic floor exercises, and review at 6–12 weeks for continence.
6. **Haematoma**: small → analgesia and ice; large or expanding → evacuation in theatre.
7. **Continuous urine leakage** after obstructed labour: suspect **VVF**. Insert an **indwelling catheter for 2–4 weeks** (small early fistulae may heal) and refer to a fistula centre.

C. Puerperal sepsis: step by step

DRUG DOSING

Puerperal sepsis / endometritis regimen (WHO MCPC, Uganda):

Drug	Dose	Route	Frequency	Duration / notes
Ampicillin	**2 g**	IV	Every 6 h	Until afebrile for **48 h**
Gentamicin	**5 mg/kg**	IV	**Once daily**	Monitor renal function; avoid prolonged use
Metronidazole	**500 mg**	IV	Every 8 h	Anaerobic cover
Then oral step-down	Amoxicillin 500 mg 8-hourly + metronidazole 400 mg 8-hourly	Oral	—	To complete **7 days** in total (if the bacteraemia or source warrants it)
Alternative (common in Ugandan hospitals)	**Ceftriaxone 1–2 g IV daily + metronidazole 500 mg IV 8-hourly**	IV	—	Per local antibiogram
Penicillin allergy	Clindamycin 600–900 mg IV 8-hourly + gentamicin	IV	—	

Algorithm (sepsis in a postnatal woman):

1. **Recognise**: fever or hypothermia, tachycardia >100, RR ≥22, altered mentation, SBP ≤100 (qSOFA-style), plus a likely source.
2. **Resuscitate**: oxygen; IV access; **crystalloid bolus 500 mL–1 L** (reassess after each; avoid overload, especially with pre-eclampsia); catheterise for urine output.
3. **Take cultures** (blood, HVS, urine) and do a **malaria test** before or at the same time as antibiotics. **Don't delay antibiotics**: give them **within 1 hour**.
4. **Give IV antibiotics** (table above).
5. **Source control**: **evacuate retained products** (MVA, gentle, after antibiotics have started); **drain abscesses**; debride infected wounds; laparotomy for peritonitis or pelvic abscess; **hysterectomy** for a necrotic, gangrenous or perforated uterus that isn't responding.
6. Treat **malaria** if positive (IV/IM **artesunate** for severe malaria, 2.4 mg/kg at 0, 12 and 24 h, then daily; oral ACT for uncomplicated).
7. **Supportive**: paracetamol, thromboprophylaxis, nutrition, breastfeeding support (continue, or express if very ill; most antibiotics above are compatible).
8. **Escalate**: persistent hypotension despite fluids → vasopressors and HDU/ICU at a RRH/NRH.
9. **If no improvement in 48–72 h**: re-examine, ultrasound for collections, review cultures, consider septic pelvic thrombophlebitis (add heparin), necrotising fasciitis, or a non-genital source.

> **UGANDA CONTEXT**
> At a **HC III**: give the **first dose of IV/IM antibiotics** (ampicillin + gentamicin + metronidazole; or IM ceftriaxone if that is what's available), IV fluids and paracetamol, test for malaria, and **refer** to a HC IV or hospital for continued care and source control (MVA, drainage, surgery). Remember that MVA for retained products is a **BEmONC** function the HC III can do.

D. Breastfeeding support and problems

Support (WHO/UNICEF Baby-Friendly): skin-to-skin and **initiation within 1 hour**; **rooming-in**; **feed on demand** (8–12 times per 24 h); **exclusive breastfeeding for 6 months**, then continue with complementary foods to 2 years or more; no teats or pre-lacteal feeds.

Good attachment (4 signs): mouth **wide open**; **lower lip turned out**; **chin touching the breast**; **more areola visible above** the mouth than below. Effective suckling: slow deep sucks with pauses, audible swallowing. Positioning: baby's head and body in line, facing the breast, close to the mother, whole body supported.

Problem	Cause	Management
Sore / cracked nipples	**Poor attachment** (most common)	Correct attachment; apply expressed hindmilk to the nipple and air-dry; continue feeding; start on the less sore side
Engorgement (days 3–5)	Milk comes in and isn't removed effectively	**Frequent effective feeding**, express to soften the areola before latching, warm compress before and **cold compress after** feeds, paracetamol/ibuprofen, supportive bra
Blocked duct	Localised milk stasis (tight clothing, missed feeds)	Tender lump, no fever: keep feeding (start with the affected side), gentle massage towards the nipple, vary positions

Problem	Cause	Management
Mastitis	Milk stasis ± infection (*Staph. aureus*), days 5–30	**Continue breastfeeding or expressing** (it is safe for the baby and essential for recovery); rest, fluids, analgesia; if symptoms last more than 12–24 h, there are fissures, or she is unwell: **cloxacillin/flucloxacillin 500 mg orally every 6 h for 10–14 days** (**erythromycin 250–500 mg 6-hourly** if penicillin-allergic). If no improvement in 48 h, suspect an **abscess**
Breast abscess	Undertreated mastitis	Fluctuant tender mass: **needle aspiration** (ultrasound-guided if possible) or **incision and drainage** (radial incision away from the areola), antibiotics; feed from the other breast and express the affected one; resume on the affected side once the drainage is away from the nipple
Nipple/oral thrush	Candida	Shiny red nipple, burning pain; white plaques in the baby's mouth. **Treat both**: miconazole cream to the nipple after feeds, nystatin oral suspension 1 mL four times daily for the baby, for 7 days
"Not enough milk"	Usually perceived; true low supply from infrequent feeding, poor attachment, supplements, retained placenta, Sheehan's	Check baby's weight gain and urine output (≥6 wet nappies/day after day 5); correct attachment; feed more often; avoid formula/water; skin-to-skin; treat the cause
Inverted/flat nipples	Anatomical	Attachment to the breast (not the nipple); syringe eversion; expressing

Lactation suppression (stillbirth, neonatal death, mother's choice where appropriate): a firm bra, cold compresses, analgesia, avoid stimulation; **cabergoline 1 mg single dose** within 24 h of birth (avoid in hypertension/pre-eclampsia). **Bromocriptine is no longer recommended** (risk of stroke, MI, seizures).

Drugs and breastfeeding: most common drugs are compatible (paracetamol, ibuprofen, penicillins, cephalosporins, metronidazole, gentamicin, nifedipine, methyldopa, labetalol, enalapril, **sertraline**, **TLD**, heparin/LMWH, warfarin). Avoid or use with caution: **ergometrine** (suppresses lactation), **combined oestrogen pills** (reduce supply early), high-dose aspirin, tetracyclines (long courses), amiodarone, lithium (specialist), and chemotherapy.

E. Postpartum family planning (PPFP)

Why: a **birth-to-pregnancy interval of at least 24 months** reduces maternal, perinatal and infant risks. Ovulation can come **before the first period**: from about **day 25** in non-breastfeeding women.

LAM: over **98% effective** only if **all 3** criteria are met:

1. **Amenorrhoea** (no vaginal bleeding after day 56)
2. **Fully or nearly fully breastfeeding**, day and night (no gap longer than about 4 h by day or 6 h at night; little or no supplementation)
3. Baby **less than 6 months** old

As soon as **any** criterion is lost, start another method.

Method, timing and WHO MEC category for breastfeeding women:

Method	When it can start postpartum	Notes
Copper IUD / LNG-IUD (postpartum IUD)	**Within 48 h** (ideally within 10 min of the placenta, including at CS), **or from 4 weeks**	Between 48 h and 4 weeks = **category 3** (higher expulsion/perforation risk). Not with puerperal sepsis (**category 4**)
Implant (Jadelle, Implanon)	**Immediately** (before discharge)	MEC 2015: category 2 under 6 weeks if breastfeeding (benefits outweigh risks); widely offered immediately in Uganda
Progestogen-only pill	**Immediately**	Take at the same time daily (within 3 h)
DMPA (injectable, Depo-Provera / Sayana Press)	MEC 2015: from **6 weeks** if breastfeeding (category 3 before 6 weeks); **immediately** if not breastfeeding	Uganda commonly gives it at the 6-week visit
Combined hormonal contraception (COC)	Breastfeeding: **not before 6 months** ideally (**category 4 under 6 weeks**, category 3 at 6 weeks to 6 months). Not breastfeeding: **not before 21 days** (VTE risk; category 4 under 21 days)	Oestrogen reduces milk supply and increases VTE risk
Tubal ligation (bilateral tubal ligation, BTL)	**Within 7 days** (or at CS), or **after 6 weeks**	Delay in 7–42 days (category D) and with sepsis, severe pre-eclampsia or major PPH until resolved. Needs **informed voluntary consent**, never conditional on care
Vasectomy	Any time	Partner option; not effective for 3 months
Condoms	Any time	Dual protection (HIV/STI)

> **UGANDA CONTEXT**
> Uganda's FP guidelines strongly promote **immediate PPFP**: counselling during ANC, then an **implant or postpartum IUD before discharge**. **Consent must be voluntary and informed**; offering a method is part of respectful care, but it must never be a condition for discharge.

F. Postpartum mood disorders

Screening: ask the **Whooley questions** ("During the past month have you often been bothered by feeling down, depressed or hopeless? … by having little interest or pleasure in doing things?") at every contact, and use the **EPDS** (10 items; ≥10 possible, ≥13 probable depression; any positive response on **item 10** needs urgent assessment).

Postpartum blues: reassurance, rest, practical support; review at 2 weeks.

Postpartum depression:

1. **Assess risk** (suicide, harm to the baby, neglect), severity, support and domestic violence.
2. **Mild–moderate**: psychosocial support and psychological therapy (e.g. **interpersonal therapy**, **problem-solving therapy**, WHO *Thinking Healthy*, group interpersonal therapy as delivered in Uganda by lay health workers), family involvement, sleep and practical help.
3. **Moderate–severe or not responding**: an antidepressant. **Sertraline 50 mg daily** (preferred in breastfeeding; titrate to 100–200 mg) or **fluoxetine 20 mg daily** (on the Uganda EML). Continue for at least **6–12 months** after remission. Amitriptyline (UCG) is an alternative, but it is dangerous in overdose.
4. **Follow up** every 2 weeks initially; refer to mental health services (e.g. the district mental health unit or **Butabika**) if severe or suicidal.

Puerperal psychosis: a psychiatric emergency

1. Ensure **safety**: never leave her alone with the baby; remove means of harm.
2. **Exclude organic causes**: sepsis, malaria, eclampsia, hypoglycaemia, electrolyte imbalance, drugs, HIV/CNS infection (check vital signs, glucose, malaria test, BP/urine).
3. **Urgent psychiatric admission** (ideally keeping mother and baby together with supervision).
4. **Antipsychotic**: e.g. **haloperidol** (UCG: 5 mg orally or IM, repeated as needed for acute agitation) or **olanzapine** 5–10 mg daily; benzodiazepine for agitation or sleep; a mood stabiliser for mania under specialist care (lithium or valproate have breastfeeding and teratogenic implications, so they need a specialist).
5. **Future pregnancies**: high recurrence risk (about 50%), especially with **bipolar disorder**; plan preventive care.

G. Other postnatal conditions to manage

- **Postpartum pre-eclampsia**: BP ≥140/90 with symptoms or proteinuria; $MgSO_4$ if severe features; antihypertensives (**nifedipine**, **methyldopa** is less preferred postpartum because it can worsen depression, **labetalol**, **enalapril**); review BP at days 3–5 and at 6 weeks.
- **VTE prophylaxis / treatment**: early mobilisation and hydration for all; LMWH (e.g. **enoxaparin 40 mg SC daily**) for high-risk women (CS plus other risk factors) for 10 days to 6 weeks; treatment doses for confirmed DVT/PE (warfarin is compatible with breastfeeding).
- **Urinary retention**: if she hasn't voided within 6 h or has a palpable bladder, **catheterise**; if the residual is more than about 500 mL, leave an indwelling catheter for 24–48 h.
- **Anaemia** (postpartum anaemia is usually defined as Hb <10 g/dL, e.g. RCOG; many Ugandan units treat below 11 g/dL): oral iron (ferrous sulphate 200 mg twice to three times daily) + folic acid for 3 months; Hb <7 g/dL or symptomatic → consider transfusion (or IV iron where available).

H. Complications

Complication	Key point
Secondary PPH	Endometritis, retained products, subinvolution (see the PPH notes)
Puerperal sepsis → septic shock, death	Leading killer in some Ugandan hospitals; group A strep can kill within hours
Pelvic abscess, septic pelvic thrombophlebitis, necrotising fasciitis	Suspect when fever persists despite antibiotics
VTE / PE	Leading direct cause of maternal death in high-income countries; underdiagnosed in Uganda
Postpartum pre-eclampsia / eclampsia, stroke	Up to 6 weeks postpartum
Peripartum cardiomyopathy	Heart failure from the last month of pregnancy to 5 months postpartum
Breast abscess, lactation failure	Poor attachment, delayed treatment
Mood disorders, suicide	Suicide is a leading cause of late maternal death
Incontinence, prolapse, fistula	Pelvic floor damage, obstructed labour
Dyspareunia	Perineal scarring, vaginal dryness during lactation
Sheehan's syndrome	Failure of lactation after major PPH
Unintended short-interval pregnancy	Missed PPFP

7. Guidelines, Policy and Law in Uganda

UGANDA CONTEXT

This topic follows the **Uganda Clinical Guidelines (UCG) 2023**, the **Essential Maternal and Newborn Clinical Care Guidelines for Uganda (MoH 2022)**, the **WHO 2022 recommendations on maternal and newborn care for a positive postnatal experience**, the **WHO Medical Eligibility Criteria for contraceptive use (5th edition, 2015)** and the Uganda family planning and consolidated HIV guidelines.

- **Postnatal stay and contacts**: at least **24 hours** in the facility after a normal birth; WHO's four contacts (**within 24 h, day 3, days 7–14, 6 weeks**), taught in Uganda as **"6 hours, 6 days, 6 weeks"** (section 2).
- **Puerperal sepsis**: first-dose IV/IM antibiotics, fluids and referral from HC III; continued care and source control (MVA, drainage, surgery) at HC IV and above (section 6C).
- **Breastfeeding**: WHO/UNICEF Baby-Friendly practices; exclusive breastfeeding for 6 months; HIV-positive mothers on ART breastfeed (Uganda consolidated HIV guidelines).
- **Postpartum family planning**: immediate PPFP is national policy (implant or postpartum IUD before discharge), with **voluntary, informed consent**; sterilisation must never be a condition of care (section 6E).
- **Mental health**: the **Mental Health Act 2018** governs admission and treatment of a woman who lacks capacity (for example, puerperal psychosis) Verify *the relevant sections on involuntary admission in the Mental Health Act 2018*. Severe cases are referred to the regional mental health unit or **Butabika**.
- **Maternal deaths** up to 42 days after birth are notifiable and reviewed under **MPDSR** (see the EmONC and safe motherhood topic).

UGANDA CONTEXT

Where Ugandan practice and WHO/international guidance differ:

- **Postnatal schedule**: "6 hours, 6 days, 6 weeks" (Uganda) vs within 24 h, day 3, days 7–14 and 6 weeks (WHO 2022). The Ugandan day-6 visit falls just before WHO's day 7–14 window; follow the MoH card in practice and quote WHO's four contacts in exams.
- **Postpartum anaemia**: usually defined as Hb below 10 g/dL (e.g. RCOG); many Ugandan units treat below 11 g/dL.
- **DMPA while breastfeeding**: MEC 2015 category 3 before 6 weeks, so Uganda commonly gives it at the 6-week visit Verify *whether the current Uganda FP guidelines or a newer WHO MEC edition allow DMPA before 6 weeks in breastfeeding women*.
- **Methyldopa** is still widely prescribed in Uganda but is less preferred postpartum because it can worsen depression; nifedipine, labetalol or enalapril are preferred.

8. Key Points

High-yield points

- **The woman is still a patient after the baby is born**: most maternal deaths happen after birth, mostly in the first 24–48 hours.
- **Puerperium** = placenta to 6 weeks (42 days); the fundus falls a finger-breadth a day and is impalpable by day 10–14.
- **Lochia**: rubra (1–4), serosa (4–10), alba (10 to 4–6 weeks); offensive or persistently red = infection or retained products.
- **Every contact**: vitals, then BUBBLE-HE, then danger signs, feeding and family planning.
- **A high, deviated fundus on day 1 is usually a full bladder.**

- **Puerperal fever: 7 Ws plus malaria**; examine every system before choosing antibiotics; antibiotics within 1 hour in sepsis.
- **Mastitis: keep breastfeeding**; cloxacillin or flucloxacillin 500 mg 6-hourly for 10–14 days.
- **LAM only while all three criteria hold**; no COC under 6 weeks if breastfeeding; PPIUD within 48 h or from 4 weeks.
- **Blues vs depression vs psychosis**: days 3–10 self-limiting vs 2 weeks or more vs an emergency in the first 2 weeks.
- **EPDS item 10** positive = same-day assessment.

Numbers & doses

Item	Value
Puerperium	Delivery of the placenta to **6 weeks (42 days)**
Uterine weight	**~1000 g** at birth → ~500 g at 1 week → ~300 g at 2 weeks → **50–100 g** at 6 weeks
Fundal descent	About **1 cm (a finger-breadth) per day**; impalpable by **day 10–14**
Lochia	Rubra **days 1–4**; serosa **~4–10**; alba **~10 to 4–6 weeks**
Postnatal stay	At least **24 h** after vaginal birth; vitals and bleeding **every 15 min for the first 2 h**
WHO contacts	**Within 24 h, day 3, days 7–14, 6 weeks** ("1-3-7-42")
First void	Within **6 h**; residual **>500 mL** → indwelling catheter **24–48 h**
Normal WBC early postpartum	Up to **20–25 × 10^9/L**
Puerperal pyrexia	**≥38 °C** in the first **14 days** (classic UK definition)
Puerperal sepsis regimen	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly**, until afebrile **48 h**, then oral to complete **~7 days**
Antibiotics in sepsis	Within **1 hour**
Mastitis	**Cloxacillin/flucloxacillin 500 mg orally 6-hourly for 10–14 days**; keep breastfeeding
Lactogenesis II	**Day 2–4** (progesterone withdrawal)
Feeds	**8–12 per 24 h**; initiate within **1 h**; exclusive for **6 months**
Adequate intake	**≥6 wet nappies/day after day 5**
LAM	**>98%** if amenorrhoea + full breastfeeding (gaps ≤4 h day, ≤6 h night) + baby **<6 months**
Ovulation (not breastfeeding)	From about **day 25**
Postpartum IUD	**Within 48 h or from 4 weeks** (48 h–4 weeks = MEC 3)
COC	Breastfeeding: **MEC 4 under 6 weeks**; not breastfeeding: **MEC 4 under 21 days**
DMPA if breastfeeding	From **6 weeks** (MEC 3 before; check current Uganda FP guidance)
Tubal ligation	**Within 7 days** or **after 6 weeks**
Birth-to-pregnancy interval	At least **24 months**
EPDS	**≥10** possible, **≥13** probable depression; **item 10** = same-day assessment
Blues / psychosis	Blues days **3–10**; psychosis about **1–2 per 1,000**, usually in the first **2 weeks**
Sertraline	**50 mg daily** (titrate to 100–200 mg)

Item	Value
Cabergoline (suppression)	**1 mg single dose** within 24 h; avoid in hypertension
Anti-D	Within **72 h** if Rh-negative with Rh-positive baby
Enoxaparin prophylaxis	**40 mg SC daily**, 10 days to 6 weeks for high-risk women
Paracetamol	**1 g every 6–8 h** (max 4 g/day)
Iron + folic acid	Continue about **3 months** after birth

Exam traps

- **"The fundus is high, so it's atony."** On day 1 a high, deviated fundus in a woman who hasn't voided is usually a **full bladder**. Catheterise.
- **"Stop breastfeeding because of mastitis."** No: **keep feeding or expressing**; stopping causes stasis and abscess.
- **"Fever on day 4 = endometritis, start antibiotics."** First **examine every system (7 Ws)** and **test for malaria**.
- **"WBC 18, so she's septic."** It can be normal early postpartum; judge the whole picture.
- **"She's tearful on day 4, so start an antidepressant."** That is **blues**; reassure and review at 2 weeks.
- **"Give COCs at the 6-week visit to a breastfeeding mother."** MEC 3 until 6 months; offer an implant, IUD, POP or DMPA instead.
- **"IUD at 2 weeks postpartum."** Wrong window: **within 48 h or from 4 weeks**.
- **Bromocriptine for lactation suppression.** No longer recommended; use **cabergoline** (not if hypertensive).

Questions seniors ask

Viva questions

1. *Define the puerperium.* Delivery of the placenta to 6 weeks (42 days).
2. *Describe the normal involution of the uterus.* Fundus at the umbilicus after birth, descending about 1 cm/day, not palpable by day 10–14; 1 kg to about 60 g by 6 weeks; by autolysis and ischaemia.
3. *Types of lochia?* Rubra, serosa, alba (with timings).
4. *Causes of puerperal pyrexia?* The 7 Ws plus malaria.
5. *Hormones of lactation?* Prolactin (production) and oxytocin (ejection); progesterone withdrawal triggers lactogenesis II.
6. *LAM criteria?* Amenorrhoea, full breastfeeding, baby under 6 months.
7. *When can a postpartum IUD be inserted?* Within 48 h or after 4 weeks.
8. *Why avoid COCs in early lactation?* They reduce milk supply and increase VTE risk (MEC 4 under 6 weeks if breastfeeding).
9. *Blues vs depression vs psychosis?* Timing, severity, duration, psychotic features, risk.
10. *WHO postnatal schedule?* Within 24 h, day 3, days 7–14, 6 weeks; stay at least 24 h.

On the ward round

What do you check at a postnatal round? Vital signs, then BUBBLE-HE: breasts, uterus, bladder, bowels, lochia, episiotomy or wound, legs, emotions; plus the baby's feeding, danger signs and family planning.

How do you know involution is normal? The fundus is firm, central and non-tender, about a finger-breadth lower each day, and impalpable by day 10–14, with lochia changing from rubra to serosa to alba and never offensive.

She has a fever on day 4. How will you find the source? The 7 Ws plus malaria: examine the uterus and lochia, wound, perineum, breasts, chest, urine and calves; send FBC, malaria test, blood culture, urine and HVS; treat the most likely source.

Her fundus is at the umbilicus on day 7 with offensive lochia. What is going on and what will you do? Subinvolution with endometritis, likely retained products: IV antibiotics, ultrasound, then gentle evacuation after antibiotics have started.

How do you tell a baby is getting enough milk? At least 6 wet nappies a day after day 5, audible swallowing, settling after feeds, and weight gain (most babies are back to birth weight by about 2 weeks).

Which contraceptive can this breastfeeding mother start before discharge? An implant, a POP, or a postpartum IUD within 48 h; not a COC. LAM only while all three criteria hold.

How do you tell blues from depression from psychosis? Blues: days 3–10, mild, self-limiting. Depression: two weeks or more, impairs functioning, screen with EPDS. Psychosis: usually the first 2 weeks, with delusions, hallucinations or confusion; an emergency.

Why must you exclude organic causes before diagnosing puerperal psychosis? Sepsis, cerebral malaria, eclampsia, hypoglycaemia and HIV-related CNS infection can all present with confusion and agitation, and they need urgent medical treatment.

Mnemonics

- **Puerperal fever: 7 Ws + M**: **W**omb, **W**ind, **W**ater, **W**ound, **W**alking (DVT), **W**eaning (breast), **W**onder drugs, and **M**alaria.
- **Lochia "R-S-A: Red, Serous, Absent colour"**: rubra (1–4), serosa (4–10), alba (10 to 6 weeks).
- **Uterus "a finger a day"**: the fundus descends one finger-breadth a day and disappears behind the symphysis by day 10–14.
- **LAM "3 As"**: **A**menorrhoea, **A**ll breastfeeding, **A**ge under 6 months.
- **Attachment "WOLF-A"**: **W**ide mouth, **O**utturned lower lip, **L**ess areola below (more above), **F**ace (chin) touching the breast, **A**udible swallowing.
- **Postnatal visits "1-3-7-42"**: day **1** (within 24 h), day **3**, days **7**–14, and day **42** (6 weeks).
- **PPIUD "48 hours or 4 weeks."**

OSCE tips

- **Breastfeeding counselling station**: observe a feed, name the **4 signs of good attachment**, and use open questions.
- **Postnatal examination station**: verbalise consent, a chaperone and hand hygiene; examine **breasts, fundus, wound/perineum, lochia and calves** (BUBBLE-HE); finish with **danger signs and FP**.
- **FP station**: explain the **LAM criteria** and the **timing of each method** clearly ("48 hours or 4 weeks" for an IUD; no COC before 6 weeks if breastfeeding).
- **Puerperal fever station**: answer with a systematic search for the source (7 Ws + malaria), the examination, the investigations and empirical treatment (section 5, Thinking like a clinician).

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define the puerperium.	From **delivery of the placenta to 6 weeks (42 days)** after birth, while the body returns to its non-pregnant state and lactation is established.
2	Normal uterine involution: fundal height and weight.	Fundus at the **umbilicus** after birth, descending about **1 cm (one finger-breadth) per day**, **not palpable abdominally by day 10–14**. Weight about 1000 g → about 500 g (1 week) → about 50–100 g (6 weeks).
3	Types and timing of lochia.	**Rubra** (red) days 1–4; **serosa** (pink-brown) about days 4–10; **alba** (yellow-white) about day 10 to 4–6 weeks.
4	WHO 2022 postnatal contact schedule.	Stay **at least 24 h** after vaginal birth; at least **4 contacts**: **within 24 h**, **day 3 (48–72 h)**, **days 7–14**, **6 weeks**. Uganda teaches '6 hours, 6 days, 6 weeks'.
5	Which hormone change triggers lactogenesis II ('milk coming in')?	The **fall in progesterone** after delivery of the placenta, with prolactin already high. It happens about **day 2–4**. Retained placental fragments can delay it.
6	Roles of prolactin and oxytocin in lactation.	**Prolactin** (anterior pituitary): milk **production**, released by suckling, higher at night. **Oxytocin** (posterior pituitary): **milk ejection** (myoepithelial contraction); inhibited by pain and anxiety; also causes afterpains.
7	What is the feedback inhibitor of lactation (FIL)?	A whey protein that builds up when milk stays in the breast and **reduces production locally**. It explains **'supply follows demand'**: frequent emptying keeps supply up.
8	LAM criteria.	All three: **amenorrhoea**, **fully or nearly fully breastfeeding** day and night, and a **baby under 6 months**. Then >98% effective. Once any criterion is lost, start another method.
9	Timing of postpartum IUD insertion.	**Within 48 h** of birth (ideally within 10 min of the placenta, including at CS) **or from 4 weeks**. Between 48 h and 4 weeks = MEC category 3. Puerperal sepsis = category 4.
10	Combined oral contraceptive: MEC categories postpartum.	Breastfeeding: **category 4 under 6 weeks**, category 3 from 6 weeks to 6 months. Not breastfeeding: **category 4 under 21 days** (VTE risk).
11	WHO definition of puerperal sepsis.	Genital-tract infection from **ROM/labour to day 42 postpartum** with **fever** plus ≥1 of: **pelvic pain**, **abnormal vaginal discharge**, **abnormal smell of discharge**, **delayed uterine involution**.
12	First-line antibiotics for puerperal sepsis (WHO/Uganda).	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly** until afebrile for 48 h. Plus source control (evacuate retained products, drain pus).
13	Causes of puerperal pyrexia (7 Ws + M).	**W**omb (endometritis), **W**ind (chest), **W**ater (UTI), **W**ound, **W**alking (DVT/PE), **W**eaning (breast), **W**onder drugs, and **M**alaria in Uganda.
14	Treatment of lactational mastitis.	**Keep breastfeeding or expressing** from the affected side; rest, fluids, analgesia; **cloxacillin/flucloxacillin 500 mg orally 6-hourly for 10–14 days** (erythromycin if penicillin-allergic). No improvement in 48 h → suspect an **abscess**.
15	Four signs of good breastfeeding attachment.	Mouth **wide open**; **lower lip turned out**; **chin touching the breast**; **more areola visible above** the mouth than below.

#	Question	Answer
16	Postpartum blues vs depression vs psychosis: onset and frequency.	**Blues**: days 3–5, resolves by about 2 weeks, very common (up to 50–80%). **Depression**: symptoms lasting 2 weeks or more, weeks to months postpartum, about 10–15% or more. **Psychosis**: usually within the **first 2 weeks**, 1–2 per 1000, an **emergency**.
17	EPDS: cut-offs and the key item.	Edinburgh Postnatal Depression Scale, 10 items: **≥10 possible**, **≥13 probable** depression. **Item 10 (self-harm)**: any positive answer needs **same-day** assessment.
18	Preferred antidepressant in breastfeeding.	**Sertraline** 50 mg daily (titrate), with low milk transfer. Fluoxetine 20 mg (Uganda EML) is an acceptable alternative. Combine with psychological therapy.
19	Why are women at high VTE risk in the puerperium?	Clotting factors and fibrinogen stay raised (**hypercoagulable for about 6 weeks**), plus venous stasis, vessel injury at delivery/CS, immobility and dehydration (Virchow's triad).
20	When does ovulation return in a non-breastfeeding woman, and what does that mean?	As early as **day 25** (mean about 45 days), **before the first period**. She needs contraception from about **3 weeks** postpartum.
21	Lactation suppression drug of choice and its caution.	**Cabergoline 1 mg orally once** within 24 h of birth. **Avoid in hypertension/pre-eclampsia.** Bromocriptine is no longer recommended (stroke, MI, seizures).
22	BUBBLE-HE: the postnatal examination.	**B**reasts, **U**terus (fundal height, tone, tenderness), **B**ladder (voided within 6 h?), **B**owels, **L**ochia (look and **smell**), **E**pisiotomy/CS wound, **H** legs/calves (DVT), **E**motions (mood, bonding). Plus temperature, pulse and BP every time.
23	Day 1: fundus above the umbilicus, deviated to the right, no urine passed in 6 h. Diagnosis and management?	**Urinary retention** (a full bladder pushing the uterus up and aside), not atony. **Catheterise**; if the residual is **>500 mL**, leave an **indwelling catheter for 24–48 h**, then a trial without catheter.
24	Is a WBC of 18 × 10^9/L on day 2 postpartum diagnostic of sepsis?	**No.** A WBC of up to about **20–25 × 10^9/L** can be normal in the first days after birth. Interpret it with the clinical picture (fever, tachycardia, RR, source) and the trend.
25	Postnatal sepsis screen: what features trigger treatment within 1 hour?	Fever ≥38 °C or low temperature, **pulse >100**, **RR ≥22**, **SBP ≤100**, **altered mentation**, with a likely source. Take cultures and a **malaria test**, then **IV ampicillin + gentamicin + metronidazole within 1 h**, fluids, and look for the source (retained products, abscess, wound).
26	How do you know a breastfed baby is getting enough milk?	**≥6 wet nappies a day after day 5**, audible swallowing, settles after feeds, and **weight gain** (most babies regain birth weight by about 2 weeks). Check attachment if in doubt.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 26-year-old para 1 is reviewed on day 5 after SVD. She is afebrile, breastfeeding well, and the lochia is pink-brown and not offensive. The fundus is palpable 4–5 finger-breadths below the umbilicus and is firm and non-tender. What is the MOST appropriate interpretation?

A Subinvolution requiring ultrasound
B Retained products requiring MVA
C Endometritis requiring antibiotics
D Normal involution and normal lochia serosa
E Full bladder requiring catheterisation

2. A 30-year-old para 3 is readmitted on day 4 after an emergency CS for obstructed labour with membranes ruptured for 22 hours. Temperature 39 °C, pulse 118/min, BP 102/64 mmHg. The uterus is tender and bulky and the lochia is foul-smelling. Malaria RDT is negative. What is the MOST appropriate initial management?

A IV ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily + metronidazole 500 mg 8-hourly, IV fluids, cultures, and ultrasound for retained products
B Oral amoxicillin 500 mg 8-hourly and review in 1 week
C Artemether-lumefantrine because malaria is common
D Immediate hysterectomy
E Paracetamol only and observe for 24 hours

3. On day 10 after delivery, a breastfeeding mother presents with a painful, red, hot, wedge-shaped area on the right breast, fever of 38.6 °C and flu-like symptoms for 36 hours. There is no fluctuant mass. What is the BEST management?

A Stop breastfeeding from both breasts
B Incision and drainage
C Stop breastfeeding from the right breast only and give paracetamol
D Bromocriptine to suppress lactation
E Continue breastfeeding/expressing from both breasts, analgesia, and oral cloxacillin 500 mg 6-hourly for 10–14 days

4. A 22-year-old para 1, day 6 after SVD, is fully breastfeeding and has no menses. She wants to avoid pregnancy for at least 3 years and is worried about daily pills. Which option is MOST appropriate to offer now?

A Combined oral contraceptive pill
B Copper IUD insertion today
C Contraceptive implant today
D Rely on LAM for 2 years
E Tubal ligation today

5. A 28-year-old on day 4 postpartum is tearful, anxious and irritable, cries when her baby cries, and says she feels overwhelmed. She is sleeping when the baby sleeps, eating, and caring for the baby well. She has no thoughts of self-harm. What is the MOST appropriate management?

A Start sertraline 50 mg daily
B Reassurance, practical support and rest, with review at 2 weeks
C Urgent psychiatric admission
D Start haloperidol
E Separate mother and baby

6. A 25-year-old primipara on day 6 postpartum is brought by her husband. She has not slept for 3 days, is talking rapidly, believes the baby has been 'switched by witches', and was found holding the baby near the fire. She has a history of a manic episode 2 years ago. Temperature 36.8 °C, glucose normal, malaria RDT negative, BP normal. What is the MOST appropriate immediate action? Advanced

A Reassure the family that this is postpartum blues
B Ensure the safety of mother and baby (constant supervision), urgent psychiatric admission and start an antipsychotic
C Refer to a traditional healer as requested by the family
D Discharge with fluoxetine 20 mg and review in 2 weeks
E Give diazepam and send home

7. A mother on day 3 complains that both breasts are hard, swollen and painful, and the baby is struggling to latch. Temperature 37.8 °C. There is no redness or focal lump. What is the MOST appropriate advice?

A Stop breastfeeding for 48 hours to rest the breasts
B Start flucloxacillin
C Give formula feeds until the swelling settles
D Express a little milk to soften the areola before latching, feed frequently on demand, cold compresses after feeds and analgesia
E Bind the breasts tightly

8. A 32-year-old para 2 with mitral stenosis, day 1 after SVD, develops sudden breathlessness, orthopnoea and bilateral basal crackles. Which physiological change of the early puerperium BEST explains this timing? Advanced

A Autotransfusion from the contracting uterus and relief of caval compression increase venous return and cardiac output immediately after delivery
B Fall in plasma volume from diuresis on day 1
C Reduced cardiac output immediately after delivery
D Anaemia from blood loss reducing preload
E Increased progesterone levels after delivery

9. A breastfeeding mother, 8 weeks postpartum, asks whether she can rely on breastfeeding for contraception. She breastfeeds exclusively day and night, but her menses returned last week. What is the BEST advice?

A Yes, LAM is reliable until 6 months
B Yes, LAM is reliable as long as she breastfeeds
C No, LAM is no longer reliable because menses have returned; start another method
D Only if she gives water in addition
E LAM is reliable until 2 years

10. A woman on day 5 after SVD has fever of 38.7 °C, headache, chills and joint pains. The uterus is well involuted and non-tender, the lochia is normal, the breasts and perineum are normal, and urinalysis is normal. She lives in Lira. What is the MOST important next investigation?

A Pelvic ultrasound
B High vaginal swab
C Chest X-ray
D Doppler ultrasound of the legs
E Malaria RDT or blood smear

11. A para 5 is referred on day 3 after a 2-day obstructed labour at home, ending in a stillbirth delivered at a HC IV. She reports continuous leakage of urine since yesterday, day and night, and is otherwise well. What is the MOST appropriate initial management? Advanced

A Insert an indwelling urethral catheter for 2–4 weeks, encourage fluids, and refer to a fistula centre
B Reassure her that it is stress incontinence that will resolve
C Immediate surgical repair of the fistula at the HC IV
D Pelvic floor exercises only
E Antibiotics for UTI and discharge

12. A 35-year-old para 4 who had an emergency CS 10 days ago presents with a painful, swollen left calf and mild breathlessness. Pulse 104/min, SpO_2 95%. What is the MOST likely diagnosis?

A Cellulitis
B Puerperal sepsis
C Muscle strain from walking
D Deep vein thrombosis with possible pulmonary embolism
E Peripartum cardiomyopathy

13. A 24-year-old HIV-positive mother on TLD with a suppressed viral load asks how to feed her 2-week-old baby. What is the BEST advice according to WHO and Uganda guidance?

A Formula feed only to eliminate transmission
B Mixed feeding with breast milk and formula from birth
C Breastfeed only after the baby is 6 months old
D Stop ART while breastfeeding to protect the baby
E Exclusive breastfeeding for 6 months, then continue breastfeeding with complementary foods, while staying on ART with viral-load monitoring and infant prophylaxis

14. Two weeks after starting cloxacillin for mastitis, a mother has a persistent 4 cm tender, fluctuant lump in the upper outer quadrant of the left breast with fever. What is the MOST appropriate management?

A Continue the same antibiotic for another 2 weeks
B Needle aspiration (ultrasound-guided if available) or incision and drainage, continue antibiotics, and keep feeding from the other breast while expressing the affected one
C Stop breastfeeding permanently
D Mastectomy
E Warm compresses only

15. A 27-year-old para 1 attends her 6-week postnatal visit. She says she has felt low for the past month, has lost interest in things, sleeps poorly even when the baby sleeps, and feels guilty that she is a bad mother. EPDS score is 17, with 'hardly ever' on item 10 (self-harm). She is breastfeeding. What is the MOST appropriate management? Advanced

A Reassure her that it is postpartum blues

B Start lithium

C Diagnose postpartum depression; offer psychological therapy and sertraline 50 mg daily, involve family support, and arrange review in 2 weeks with a safety plan

D Advise her to stop breastfeeding before any treatment

E Admit to a psychiatric hospital immediately

Answers & explanations

1. D · Normal involution and normal lochia serosa

D is right: the fundus descends about one finger-breadth (1 cm) a day, so **4–5 finger-breadths below the umbilicus on day 5** in a firm, non-tender uterus, with normal pink-brown lochia **serosa** and no fever, is **normal involution**. A and B are wrong: subinvolution or retained products cause a bulky, soft uterus with persistent red lochia or bleeding. C is wrong: there is no fever, tenderness or offensive lochia. E is wrong: a full bladder displaces the uterus upwards and to one side, and she has no urinary symptoms.

2. A · IV ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily + metronidazole 500 mg 8-hourly, IV fluids, cultures, and ultrasound for retained products

A is right: this is **puerperal sepsis (post-CS endometritis)** with sepsis physiology. Give **triple IV antibiotics within 1 hour**, fluids, cultures, and look for retained products or a collection (ultrasound) for source control. B is wrong: oral treatment is inadequate for a septic woman. C is wrong: the RDT is negative and the clinical picture points to the uterus. D is wrong: hysterectomy is for a necrotic or unresponsive uterus, not first line. E is wrong: delaying antibiotics increases mortality.

3. E · Continue breastfeeding/expressing from both breasts, analgesia, and oral cloxacillin 500 mg 6-hourly for 10–14 days

E is right: this is **infective mastitis** (symptoms over 24 h, febrile). Treatment is **continued milk removal** (the most important measure; it is safe for the baby), analgesia and an **anti-staphylococcal antibiotic** such as cloxacillin/flucloxacillin for 10–14 days. A and C are wrong: stopping feeding causes milk stasis and increases the risk of abscess. B is wrong: there is no abscess (fluctuant mass). D is wrong: lactation suppression is unnecessary, and bromocriptine is no longer recommended.

4. C · Contraceptive implant today

C is right: an **implant** is long-acting, doesn't affect breastfeeding, and can be inserted **immediately postpartum** (MEC 2015 category 2 under 6 weeks if breastfeeding). This is widely offered in Uganda. A is wrong: COCs are **category 4** under 6 weeks in breastfeeding women. B is wrong: IUD insertion between **48 h and 4 weeks** is category 3 (expulsion/perforation risk); wait until 4 weeks. D is wrong: LAM is valid only up to 6 months and while the criteria are met. E is wrong: tubal ligation is permanent and she is 22 and wants spacing, not limiting (and from day 7 to day 42 postpartum it falls in the MEC 'delay' period).

5. B · Reassurance, practical support and rest, with review at 2 weeks

B is right: this is **postpartum blues**. Onset days 3–5, emotional lability, preserved functioning, no psychotic features or suicidality; it is self-limiting. Reassure, involve the family, and review at about 2 weeks (symptoms persisting beyond that suggest depression). A is wrong: antidepressants are not indicated for blues. C and D are wrong: there are no psychotic features. E is wrong: it is harmful to bonding and breastfeeding.

6. B · Ensure the safety of mother and baby (constant supervision), urgent psychiatric admission and start an antipsychotic

B is right: this is **puerperal psychosis** (onset in the first 2 weeks, insomnia, delusions, mania), in a woman with a history suggesting **bipolar disorder**, and there is a **risk of infanticide**. Organic causes have been screened (normal temperature, glucose, BP, negative RDT). It is an **emergency**: supervision, admission and an antipsychotic (e.g. haloperidol or olanzapine). A is wrong: blues has no psychotic features. C is wrong: it delays life-saving care. D is wrong: an antidepressant alone can worsen mania, and outpatient care is unsafe. E is wrong: it fails to address the risk.

7. D · Express a little milk to soften the areola before latching, feed frequently on demand, cold compresses after feeds and analgesia

D is right: this is **engorgement** (bilateral, day 3–5, low-grade fever) from inadequate milk removal as the milk comes in. Soften the areola to help attachment, **feed frequently**, use cold compresses and give paracetamol/ibuprofen. A and C are wrong: they worsen stasis and reduce supply (and formula undermines exclusive breastfeeding). B is wrong: there is no infection. E is wrong: binding is used for lactation suppression, not engorgement in a mother who wants to breastfeed.

8. A · Autotransfusion from the contracting uterus and relief of caval compression increase venous return and cardiac output immediately after delivery

A is right: immediately after delivery, about 500 mL of blood returns from the uteroplacental circulation and caval compression is relieved. **Preload and cardiac output rise**, which a stenotic mitral valve can't accommodate, so she develops **pulmonary oedema**. B is wrong: diuresis happens on days 2–5 and reduces volume. C is wrong: cardiac output rises, not falls. D is wrong: reduced preload would not cause pulmonary oedema. E is wrong: progesterone falls after delivery.

9. C · No, LAM is no longer reliable because menses have returned; start another method

C is right: LAM requires **all three criteria**: amenorrhoea, full breastfeeding, baby under 6 months. The return of menses means ovulation is resuming, so she needs another method (e.g. implant, POP, DMPA or IUD). A, B and E are wrong: they ignore the amenorrhoea criterion. D is wrong: supplementing with water also breaks the 'full breastfeeding' criterion and is not recommended for the baby.

10. E · Malaria RDT or blood smear

E is right: in a malaria-endemic area, fever with headache, chills and arthralgia and **no genital, breast, wound or urinary source** is **malaria** until proven otherwise; malaria is more severe in pregnancy and the puerperium. A and B are less useful with a normal uterus and lochia. C is wrong: there are no chest symptoms. D is wrong: there are no leg signs.

11. A · Insert an indwelling urethral catheter for 2–4 weeks, encourage fluids, and refer to a fistula centre

A is right: continuous leakage after prolonged obstructed labour suggests an **obstetric vesicovaginal fistula** from pressure necrosis. **Early continuous bladder drainage** for 2–4 weeks may allow small fistulae to heal, and she should be referred to a fistula specialist. B and D are wrong: stress incontinence is leakage with coughing, not continuous. C is wrong: definitive repair is usually delayed until the tissues are healthy (about 3 months) and done by trained surgeons. E is wrong: it misses the diagnosis.

12. D · Deep vein thrombosis with possible pulmonary embolism

D is right: the puerperium (especially after CS) is **hypercoagulable**. A unilateral swollen painful calf with tachycardia and dyspnoea suggests **DVT with possible PE**. Start treatment-dose LMWH while arranging Doppler ± CTPA. A is wrong: cellulitis wouldn't explain the dyspnoea and tachycardia. B is wrong: there are no genital-tract symptoms. C is wrong: a strain doesn't cause swelling with tachycardia. E is wrong: cardiomyopathy causes bilateral oedema, orthopnoea and heart failure signs, not a unilateral calf.

13. E · Exclusive breastfeeding for 6 months, then continue breastfeeding with complementary foods, while staying on ART with viral-load monitoring and infant prophylaxis

E is right: with maternal ART and viral suppression, **exclusive breastfeeding** for 6 months, then continued breastfeeding with complementary foods, gives the best infant survival and a very low transmission risk; the baby also receives prophylaxis per the national guideline. A is wrong: in Uganda, formula carries risks (diarrhoea, pneumonia, cost) and is not recommended routinely. B is wrong: **mixed feeding in early infancy increases transmission risk**. C is nonsensical. D is wrong: ART must continue to keep viral load suppressed.

14. B · Needle aspiration (ultrasound-guided if available) or incision and drainage, continue antibiotics, and keep feeding from the other breast while expressing the affected one

B is right: a fluctuant tender mass after mastitis is a **breast abscess**. It needs **drainage** (aspiration or I&D with a radial incision away from the areola) plus antibiotics. Breastfeeding continues from the unaffected breast, and the affected breast is expressed until feeding from it can resume. A and E are wrong: pus needs drainage. C is wrong: stopping causes more stasis and deprives the baby. D is wrong: it is grossly excessive.

15. C · Diagnose postpartum depression; offer psychological therapy and sertraline 50 mg daily, involve family support, and arrange review in 2 weeks with a safety plan

C is right: symptoms for more than 2 weeks with anhedonia, sleep disturbance and guilt, and an EPDS of 17 (probable depression), mean **moderate–severe postpartum depression**. Treat with **psychological therapy plus an SSRI**; **sertraline** is preferred in breastfeeding. Arrange close follow-up and a safety plan, because she has some self-harm ideation ('hardly ever'). A is wrong: blues resolves by 2 weeks. B is wrong: lithium isn't first line for depression and needs specialist care. D is wrong: sertraline is compatible with breastfeeding. E is wrong: admission is for high suicide risk, psychosis or inability to care for herself or the baby.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Fever on day 4 after caesarean section

At Mengo Hospital, a 23-year-old para 1 is on day 4 after an emergency CS for obstructed labour. She laboured for 20 hours at a HC III before referral, with membranes ruptured for 24 hours and many vaginal examinations. Today she has a temperature of 38.9 °C, pulse 116, RR 22, BP 104/66.

1. What focused history will you take and why?

- **Abdominal/pelvic pain, lochia** (amount, smell): endometritis
- **Wound pain or discharge**: wound infection
- **Dysuria, frequency, loin pain**: UTI/pyelonephritis (she was catheterised)
- **Cough, breathlessness, chest pain**: chest infection (spinal or GA?), PE
- **Breast pain**: engorgement/mastitis
- **Calf pain**: DVT
- **Headache, chills, joint pains**: **malaria**
- **HIV status/ART**; drug history (antibiotics given at CS?)
- Breastfeeding and the baby's condition

2. What examination findings would you look for, and what do you find?

Look for: general (toxic appearance, pallor, hydration), chest, **breasts**, **abdomen: fundal height and tenderness**, **wound**, **lochia** (look and smell), perineum, **calves**, renal angles. Findings: tender, **bulky uterus 2 cm above the umbilicus** (subinvolution), **foul-smelling lochia**, the wound is clean and dry, breasts soft, chest clear, calves normal.

3. What is your diagnosis, differentials and investigations?

Puerperal sepsis: post-CS endometritis (risk factors: prolonged labour, prolonged ROM, multiple VEs, emergency CS). Differentials: retained products, pelvic abscess or infected haematoma, UTI, malaria, wound infection (early), chest infection. Investigations: **FBC**, **blood cultures**, **HVS**, **malaria RDT**, **urinalysis/MSU**, **creatinine and lactate**, **pelvic ultrasound** (retained products, collection).

4. Outline your management.

1. **IV access, fluids** (500 mL–1 L crystalloid bolus, reassess), catheter for urine output.
2. **Antibiotics within 1 hour**: **ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly** (or ceftriaxone + metronidazole per local protocol).
3. **Source control**: if the scan shows retained products → gentle evacuation after antibiotics have started; drain any collection.
4. **Paracetamol**; **thromboprophylaxis** (LMWH) once not bleeding; mobilise.
5. **Continue breastfeeding** (these antibiotics are compatible).
6. Monitor vital signs 4-hourly (MEOWS), fluid balance.
7. Continue IV until **afebrile for 48 h**, then oral to complete 7 days.

5. On day 3 of antibiotics she is still spiking fevers of 39 °C but looks well between spikes. The ultrasound shows no collection. What would you consider?

> **Septic pelvic thrombophlebitis** (spiking fevers despite antibiotics, no source; consider adding **therapeutic heparin**), a **resistant organism** (check cultures; escalate, e.g. to meropenem where available), **wound abscess** developing, **malaria** (repeat the test), **drug fever**, **breast abscess**, pyelonephritis, and HIV-related infections (TB). Re-examine carefully and involve a senior.

> **Take-away:** Puerperal fever needs a systematic search (7 Ws + malaria), antibiotics within an hour, and source control. If she isn't improving by 72 hours, look for pus, clots and wrong diagnoses.

Case 2: The discharge postnatal check and family planning

You are asked to discharge a 20-year-old para 2 at a HC IV, 24 hours after an uncomplicated SVD of a healthy 3.2 kg boy. Her first child is 13 months old. She is HIV-negative (tested in labour) and breastfeeding.

1. What will you check before discharge (mother)?

> Vital signs (**temperature, pulse, BP**: pre-eclampsia can appear postpartum), pallor/Hb if there was significant bleeding, **fundus** (firm, at or below the umbilicus), **lochia** (amount, no clots or smell), **perineum**, **passed urine** and bowels, **breasts and a breastfeeding observation** (attachment), **calves**, **mood**, anti-D if Rh-negative, iron/folate supply, and HIV retesting plan.

2. What danger signs will you teach her to return with?

> Heavy bleeding (more than 1 pad an hour, large clots); fever or chills; foul-smelling discharge; severe abdominal pain; **severe headache, blurred vision, fits**; difficulty breathing or chest pain; painful swollen leg; breast redness or pain; worsening perineal pain or discharge; feeling very low, not coping, or thoughts of harming herself or the baby. Plus newborn danger signs (poor feeding, fever, fast breathing, jaundice, convulsions, cord infection).

3. Her births are only 13 months apart. How will you counsel her on family planning?

> Explain the benefit of a **birth-to-pregnancy interval of at least 24 months** (lower risk of anaemia, preterm birth, low birth weight, maternal death). Explain that ovulation can return **before the first period**. Discuss options using WHO MEC for a breastfeeding woman:
>
> - **LAM** (if amenorrhoeic, fully breastfeeding, baby under 6 months), with a plan to switch
> - **Implant** or **POP today**
> - **PPIUD within 48 h** (possible today) or **from 4 weeks**
> - **DMPA from 6 weeks**
> - **Avoid COCs** until 6 months
> - Condoms for dual protection Respect her **voluntary, informed choice**; involve her partner if she wishes.

4. She chooses an implant. What else will you arrange for follow-up?

> Insert the implant (consent, aseptic technique, check it is palpable, give a card). Book postnatal contacts on **day 3** and at **days 7–14** (or VHT home visit), and at **6 weeks** (BP, FP review, VIA screening if due, **EPDS/mood**, baby's immunisations and growth). Continue **iron and folic acid** for 3 months, **exclusive breastfeeding**, and **HIV retest** per the national schedule while breastfeeding.

> **Take-away:** A discharge check is a clinical examination plus counselling: danger signs, breastfeeding and family planning all have to fit into that one conversation.

Case 3: A low mother at the 6-week visit

A 29-year-old para 3 attends the 6-week postnatal clinic at a regional referral hospital. The midwife notes she is withdrawn and tearful. Her husband left 2 months before the birth, and her last baby died in the neonatal period 2 years ago.

1. How will you screen and assess her?

> Build rapport in private. Ask the **Whooley questions** and complete the **EPDS**. Explore **duration** (more than 2 weeks?), mood, anhedonia, sleep, appetite, energy, guilt, concentration, **bonding and care of the baby**, **thoughts of self-harm or harm to the baby** (ask directly), **psychotic symptoms**, substance use, **domestic violence**, social support and finances. Look for organic contributors: **anaemia, hypothyroidism, HIV**, chronic pain.

2. Her EPDS is 19; she has had passive thoughts that 'the baby would be better off without me' but no plan. No psychotic symptoms. Diagnosis and risk?

> **Moderate–severe postpartum depression** with **passive suicidal ideation**, and risk factors of loss of partner support, previous neonatal death (grief) and poverty. Risk: **moderate** (no plan or intent, but she needs a safety plan and close follow-up). Check Hb and TSH.

3. Outline management.

> 1. **Safety plan**: identify a trusted person (sister, mother), remove means of harm, give emergency contacts, and agree she will return if thoughts worsen.
> 2. **Psychological therapy**: interpersonal/problem-solving therapy, group support (e.g. community IPT-G), grief counselling.
> 3. **Antidepressant**: **sertraline 50 mg daily** (compatible with breastfeeding); review in 2 weeks; titrate; continue at least 6–12 months after remission.
> 4. **Social support**: social worker, community resources, involve family.
> 5. **Physical**: treat anaemia, support breastfeeding and infant care.
> 6. **Referral** to mental health services if not improving or if risk rises; urgent admission if there is active intent or psychosis.
> 7. **Family planning** counselling.

4. What would make you admit her urgently?

> **Active suicidal intent or plan**, thoughts of **harming the baby** with intent, **psychotic features** (delusions, hallucinations), severe self-neglect or inability to care for the baby, or no safe support at home. Admit to a psychiatric unit (e.g. Butabika or the regional mental health unit), ideally keeping mother and baby together with supervision.

Take-away: Ask every postnatal mother about mood and self-harm directly. Postpartum depression is common and treatable, and sertraline plus talking therapy is safe with breastfeeding.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.